

Math 54 Discussion Worksheet

Name: _____ ID: _____

Problem 1

Let A be an $m \times n$ matrix. Prove that every vector $x \in \mathbb{R}^n$ can be written as $x = p + u$ where p is in $\text{Row}(A)$ and u is in $\text{Null}(A)$. Also, show that if the equation $Ax = b$ is consistent, then there is a unique p in $\text{Row}(A)$ so that $Ap = b$.

Problem 2

Suppose that $A = QR$ where Q is $m \times n$ and R is $n \times n$. Show that if the columns of A are linearly independent, then R must be invertible. (Hint: Consider the equation $Rx = 0$.)

Problem 3

Let A be an $m \times n$ matrix.

- a) Show that $\text{Null}(A) = \text{Null}(A^T A)$ by first showing that $Ax = 0$ implies $A^T Ax = 0$, then by showing that $A^T Ax = 0$ implies $x^T A^T Ax = 0$ and concluding that $Ax = 0$.
- b) Show that $A^T A$ is invertible (even though A may not be) iff the columns of A are linearly independent.
- c) Prove that $\text{rank}(A^T A) = \text{rank}(A)$.